src/flightCompV2/telemetry.cpp

```cpp
/*
* cpp file ARC-S telemetry printouts
 *
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 */
#include "Telemetry.h"
#include "servos.h"
extern craftState vehicle;

void Telemetry::serialIMU()
{
    // CSV format: ACC_X,ACC_Y,ACC_Z
    Serial.print(vehicle.accel_g.x, 3); Serial.print(",");
    Serial.print(vehicle.accel_g.y, 3); Serial.print(",");
    Serial.println(vehicle.accel_g.z, 3);
    Serial.flush(); // ensure buffer is sent
}

void Telemetry::serialIMUDeadReckoning()
{
    // CSV format: DR_X,DR_Y,DR_Z
    Serial.print(vehicle.deadReckoningPosition_m.x, 3); Serial.print(",");
    Serial.print(vehicle.deadReckoningPosition_m.y, 3); Serial.print(",");
    Serial.println(vehicle.deadReckoningPosition_m.z, 3);
    Serial.flush();
}

void Telemetry::serialGyro()
{
    // CSV format: GYRO_X,GYRO_Y,GYRO_Z
    Serial.print(vehicle.gyro_dps.x, 2); Serial.print(",");
    Serial.print(vehicle.gyro_dps.y, 2); Serial.print(",");
    Serial.println(vehicle.gyro_dps.z, 2);
    Serial.flush();
}

void Telemetry::serialGyroDeadReckoning()
{
    // CSV format: GP_X,GP_Y,GP_Z
    Serial.print(vehicle.gyroAngle_deg.x, 2); Serial.print(",");
    Serial.print(vehicle.gyroAngle_deg.y, 2); Serial.print(",");
    Serial.println(vehicle.gyroAngle_deg.z, 2);
    Serial.flush();
}

```

```
48 void Telemetry::serialMagnometer()
49 {
50     // CSV format: MAG_X,MAG_Y,MAG_Z
51     Serial.print(vehicle.mag_uT.x, 2); Serial.print(",");
52     Serial.print(vehicle.mag_uT.y, 2); Serial.print(",");
53     Serial.println(vehicle.mag_uT.z, 2);
54     Serial.flush();
55 }
56
57 void Telemetry::serialBarometer()
58 {
59     // CSV format: TEMP_C,PRESSURE_HPA,ALT_M
60     Serial.print(vehicle.temperature_C, 2); Serial.print(",");
61     Serial.print(vehicle.pressure_Pa, 2); Serial.print(",");
62     Serial.println(vehicle.altitude_m, 2);
63     Serial.flush();
64 }
65
66 void Telemetry::serialServoOutputs()
67 {
68     // CSV format (deg): PITCH+Y,PITCH-Y,YAW+X,YAW-X
69     // Uses the last commanded servo mix captured in applyServoMix()
70     // No flush() here: called every loop; blocking on flush when host isn't
   reading would stall the ESP32.
71     Serial.print(lastServoCommand.posY, 2); Serial.print(",");   // PITCH+Y
72     Serial.print(lastServoCommand.negY, 2); Serial.print(",");   // PITCH-Y
73     Serial.print(lastServoCommand.posX, 2); Serial.print(",");   // YAW+X
74     Serial.print(lastServoCommand.negX, 2); Serial.println();    // YAW-X
75 }
```